

# Deepfake Forensic Classification Report

**Packet ID:** 9125288b-f268-45f6-bf8c-017f237aa90e   **Date (IST):** 01-06-2026 19:42:53 IST

**Subject:** Shri Example Politician

**Media SHA-256:**  
a665a45920422f9d417e4867efdc4fb8a04a1f3fff1fa07e998e86f7f7a27ae3

Media classified as likely synthetically manipulated based on 12 forensic indicator(s), including 4 high-severity finding(s).

| Category            | Severity | Finding                                                                                                                                        | Metric                              |
|---------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| Audio-Video Sync    | critical | Audio–video lip movement is misaligned, consistent with synthetic dubbing or post-hoc voice replacement rather than naturally captured speech. | lip_sync_alignment_c<br>127.5<br>ms |
| Facial Manipulation | high     | Spatial facial inconsistencies detected across frames suggest GAN or deepfake-based facial manipulation.                                       | spatial_cnn_manipula<br>94.00%      |

|                        |        |                                                                                                                                                                                          |                                 |
|------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| 3D<br>Face<br>Geometry | medium | 3D<br>face<br>mesh<br>landmarks<br>are<br>unstable<br>across<br>the<br>temporal<br>sequence,<br>indicating<br>inconsistent<br>facial<br>geometry<br>typical<br>of<br>synthetic<br>faces. | face_mesh_landmark<br>0.0823    |
| Synthetic<br>Voice     | high   | The<br>voice<br>track<br>exhibits<br>characteristics<br>consistent<br>with<br>text-to-speech<br>or<br>neural<br>voice<br>cloning<br>rather<br>than<br>a<br>live<br>recording.            | tts_synthetic_probabi<br>89.00% |

|                    |        |                                                                                                                                               |                            |
|--------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| Voice Authenticity | medium | Pitch contour and spectrogram features do not match natural speech patterns for the apparent speaker, suggesting audio synthesis or splicing. | spectrogram_pitch_m 41.00% |
| Anti-Spoofing      | high   | Anti-spoofing neural network assigns low confidence to bona fide speech, indicating the audio is likely non-genuine or heavily processed.     | anti_spoofing_nn_co 12.00% |

|                      |        |                                                                                                                                                     |                             |
|----------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| Lighting Consistency | medium | Lighting on the face and background shifts unnaturally between frames, suggesting compositing or generative synthesis rather than a single capture. | illumination_inconsis72.00% |
| Shadow Geometry      | medium | Shadow direction and softness are inconsistent with the scene light source, a common artifact in deepfake and face-swap videos.                     | shadow_geometry_in68.00%    |

|                    |        |                                                                                                                                              |                            |
|--------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| Color Grading      | medium | Skin tone and color grading do not remain consistent across frames; color boundaries and shadows appear imperfect relative to natural video. | color_grading_anoma61.00%  |
| Temporal Stability | medium | High-frequency flicker and temporal instability detected at facial boundaries, consistent with frame-wise generative manipulation.           | temporal_flicker_sco58.00% |

|                     |        |                                                                                                                           |                                 |
|---------------------|--------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| Facial Boundaries   | medium | Visible artifacts at the jawline, hairline, or face perimeter suggest blending of a synthetic face onto an original body. | facial_boundary_artif<br>64.00% |
| Analyst Observation | medium | Visible jawline blending; lighting on face inconsistent with background.                                                  | —                               |

*This report is generated for court and authority review under the Bharatiya Sakshya Adhiniyam, 2023 evidentiary framework.*